



Customer Shopping Behavior Analysis

An End-to-End Data Analytics Project Using Python, SQL and Power BI

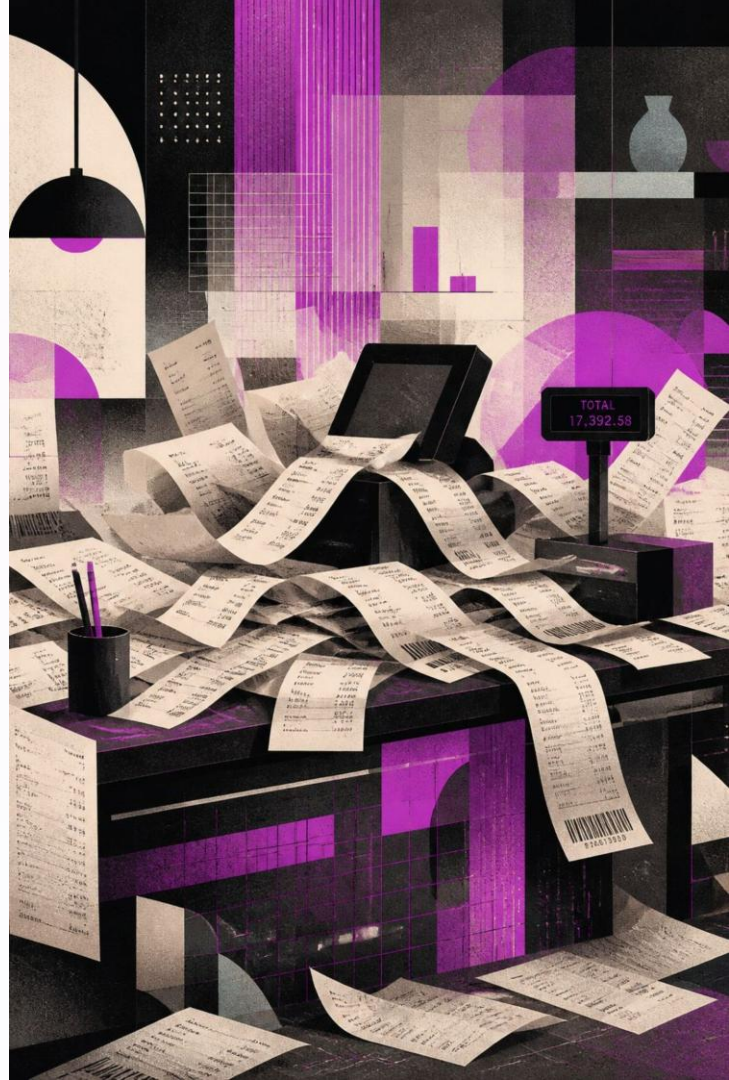
The Challenge

Data Without Direction

Retail businesses generate large transaction volumes but lack infrastructure to convert raw records into actionable intelligence.

Blind Spots in Strategy

Without clear insight into customer demographics, purchase patterns, and preferences, businesses cannot effectively target marketing or optimize product mix.



Project Scope



Dataset

3,900 customer transactions across 18 columns

Tools

Python (Pandas), MySQL, Power BI

Objectives

Clean data, execute 10 SQL analyses, build interactive dashboard

Data Cleaning & Preparation

01

Impute Missing Values

Imputed 37 missing Review Rating values using category-level median to preserve data integrity.

03

Remove Redundancy

Discovered and removed redundant column: Discount Applied = Promo Code Used (100% identical).

02

Feature Engineering

Engineered **age_group** feature via quartile segmentation: Young Adult, Adult, Middle-aged, Senior.

04

Export to MySQL

Exported cleaned 3,900 × 19 dataset to MySQL for structured SQL analysis.



Key Finding: Clothing Dominates



Clothing · 44.5%

Other Categories · 55.5%

Revenue Breakdown

Clothing generates **~\$104K revenue** — representing **44.5% of all transactions**.

1,737 clothing records vs. 250 outerwear records — a 7× volume advantage over the next closest category.



Implication: Protect inventory and prioritize marketing in this category.

Key Finding: Young Adults Lead Revenue

\$62K

Young Adult Revenue

Ages 18–30 generate the highest revenue of all age groups

1,028

Transactions

Young Adults account for the most purchase records in the dataset

11%

Revenue Edge

More revenue than the Adult segment — the next closest age group



Implication: Digital and social media targeting should focus on the 18–30 demographic.

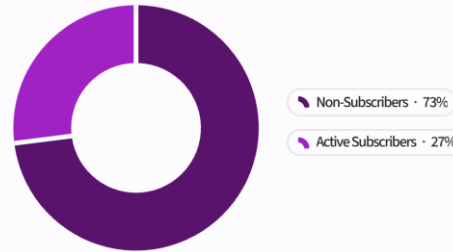
Key Finding: Subscription Opportunity

A Largely Untapped Base

73% of customers are non-subscribers — 2,847 out of 3,900 total customers.

Only **27%** hold active subscriptions (1,053 customers), leaving a significant recurring-revenue gap.

Converting just **10% of non-subscribers** adds approximately **285 recurring-revenue customers** to the base.



Key Finding: Discounts Don't Drive Spend

What the Data Shows

High-value discount users spend **at or above the \$59.76 average** — discounts are not the driver of higher spend.

Shipping type has **no material effect** on purchase amount across all 6 shipping types tested.

Strategic Implication

Replace blanket discounts with **loyalty rewards** that reinforce repeat behavior without eroding margin.

Position shipping upgrades as **perks, not upsells** — customers are indifferent to shipping type when making purchase decisions.

Dashboard Overview



KPI Cards

3.9K customers · **\$59.76** avg purchase · **3.75** avg rating — top-line metrics at a glance.



Visual Panels

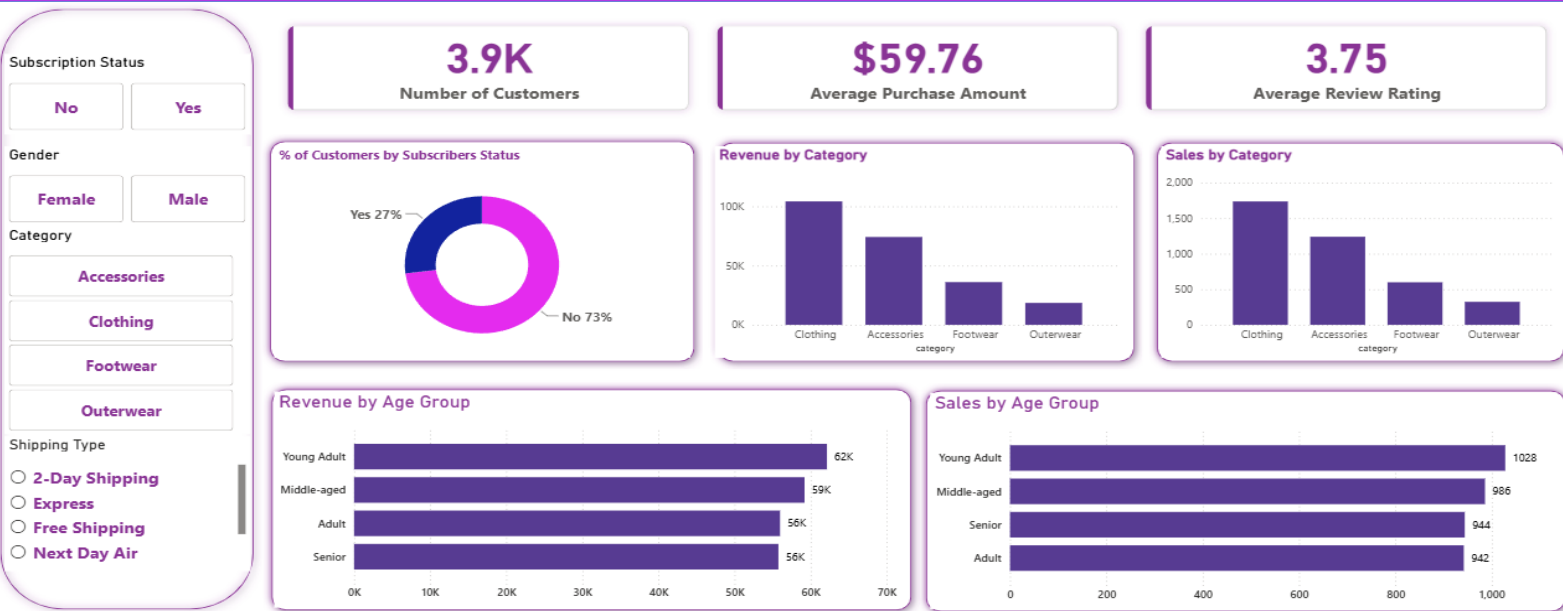
Subscription donut, revenue & sales by category, revenue & sales by age group — all in one view.



Interactive Slicers

Dynamic filters for subscription status, gender, category, and shipping type enable drill-down exploration.

Customer Behavior Dashboard



Dashboard Screenshot

Interactive Power BI dashboard delivering KPI cards, category revenue breakdowns, age group analysis, and subscription segmentation — all filterable by gender, category, and shipping type.

SQL Analysis Highlights

1

Q1 — Revenue by Gender

Male customers (68%) generate proportionally higher total revenue — informing gender-targeted campaign allocation.

2

Q3 — Top Products by Rating

Top 5 products by review rating identified as prime candidates for featured placement and promotional spotlight.

3

Q7 — Loyalty Segmentation via CTE

Customer loyalty segmentation using a Common Table Expression reveals the majority fall into the **'Loyal'** category (>10 purchases).